Object Identification and Recognition (III)

Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

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Interpretation Trees for Feature-Based Identification

The scope - objects as sets of features

Measurements (Image)

Extraction process

$$(f_1, f_2, \ldots, f_n)$$

Model Database

$$(F_1, F_2, \ldots, F_{m_1})$$

$$(F_1, F_2, \ldots, F_{m_2})$$

$$(F_1, F_2, \ldots, F_{m_3})$$

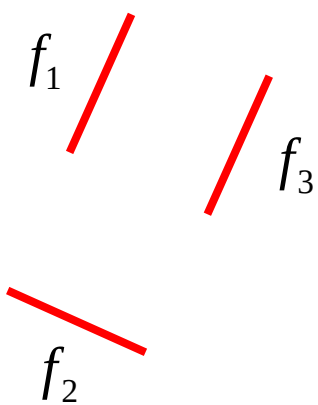
$\vdots$

$$(F_1, F_2, \ldots, F_{m_k})$$

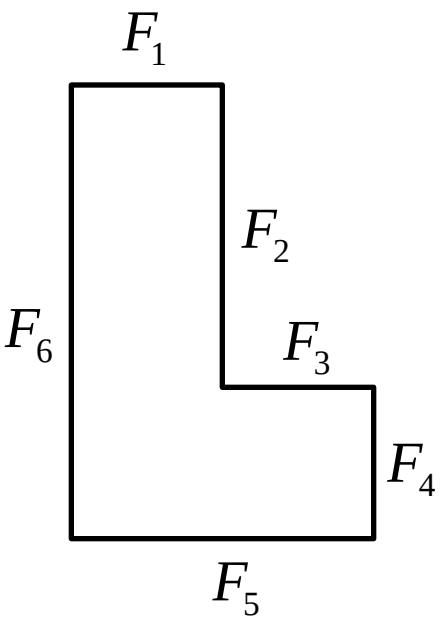
Interpretation Trees for Feature-Based Identification

Objects as local geometrical features and their inter-relationships

Measurements (Image)



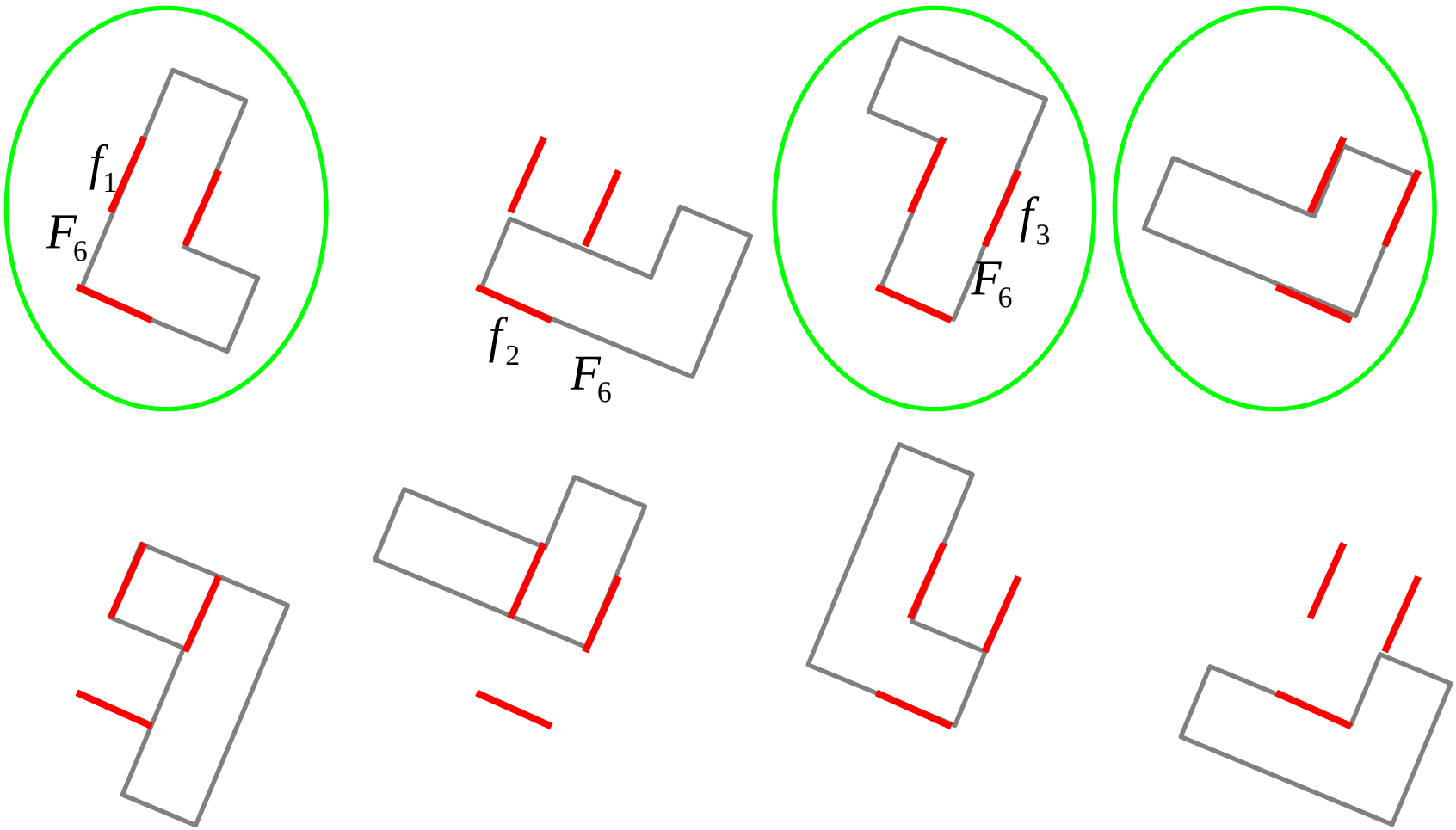
Model



Recognition = **Consistent interpretation** of the measurements in terms of the model features

Interpretation Trees for Feature-Based Identification

Recognition as search



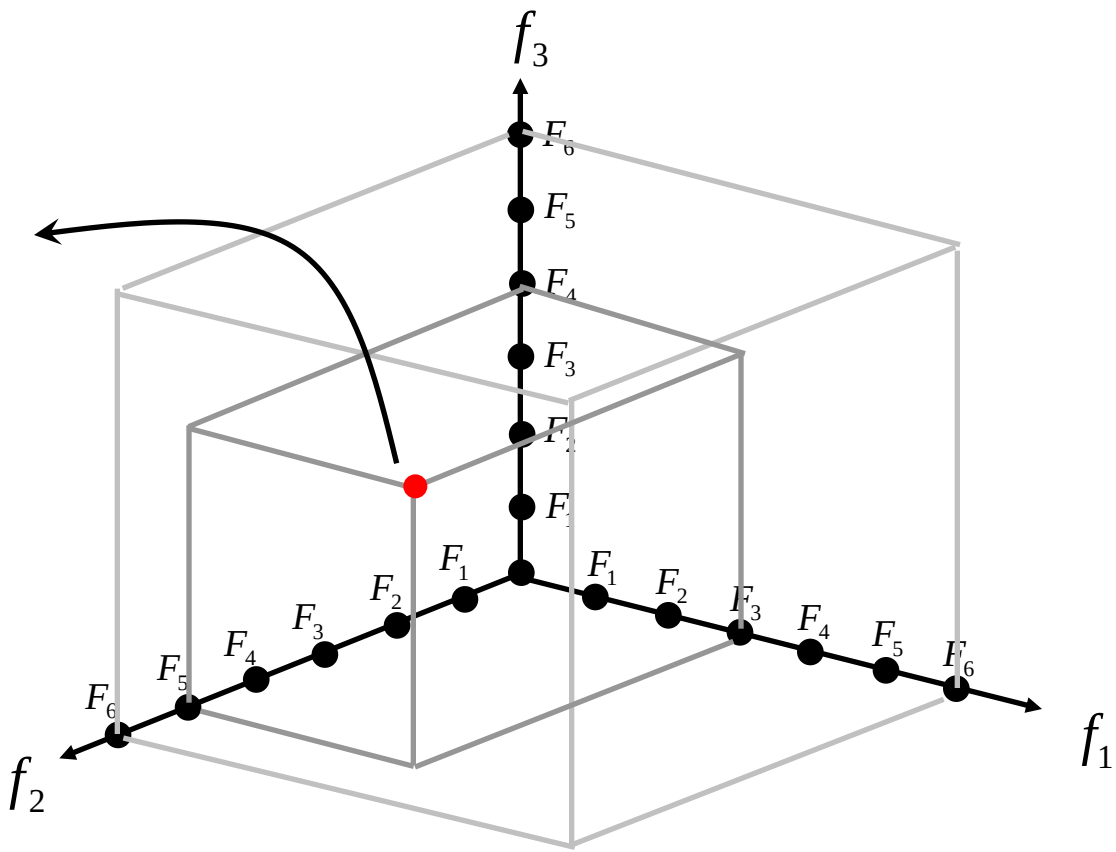
Interpretation Trees for Feature-Based Identification

The correspondence space

$f_1 \Leftrightarrow F_3$

$f_2 \Leftrightarrow F_5$

$f_3 \Leftrightarrow F_4$



Interpretation Trees for Feature-Based Identification

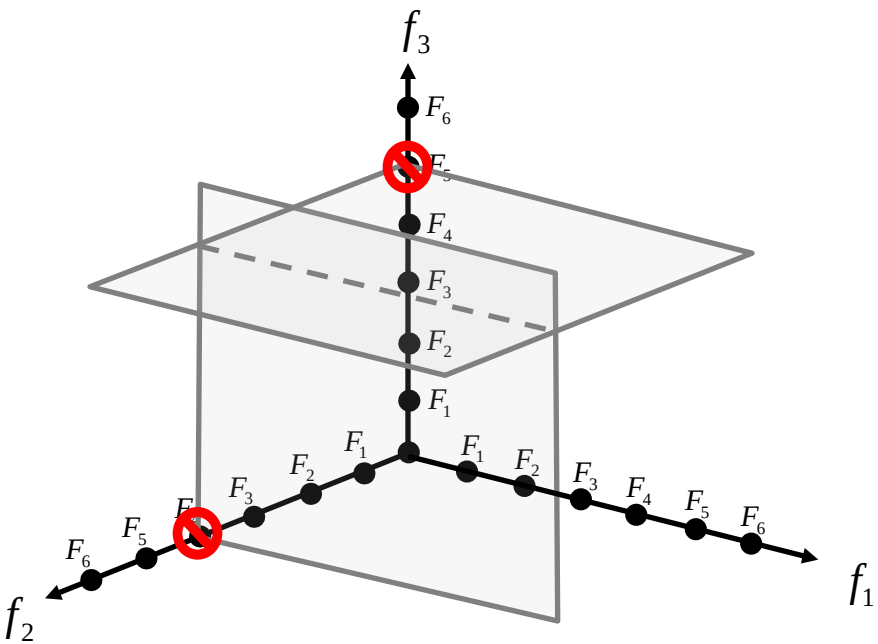
Challenges in feature matching

- Search space size
- Cost of match verification
- Occlusions (partial/missing measured features)
- Multiple objects and complex scenes
- Measurement errors in feature position and pose
- Spurious measurements

Interpretation Trees for Feature-Based Identification

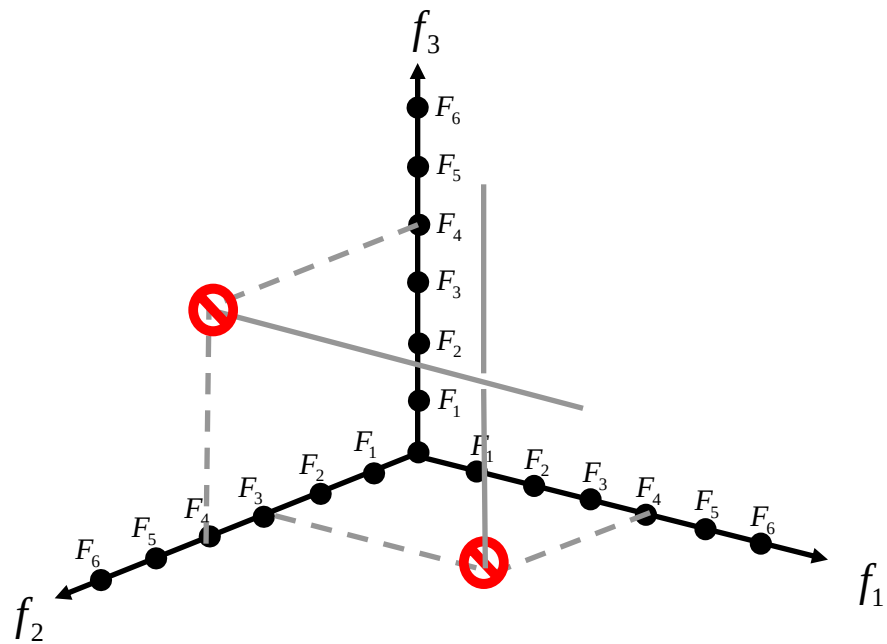
The use of constraints

Unary constraints



$$U_k(f_i, F_p) \in \{True, False\}$$

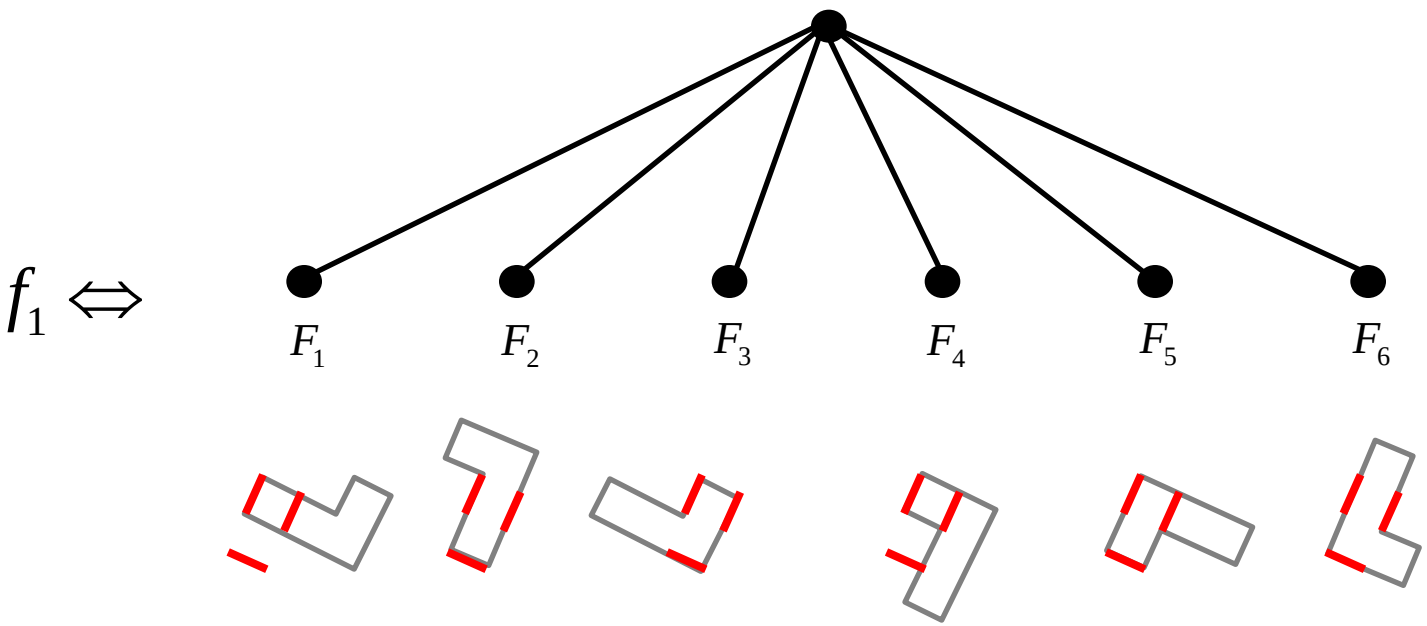
Binary constraints



$$B_k(f_i, F_p; f_j, F_q) \in \{True, False\}$$

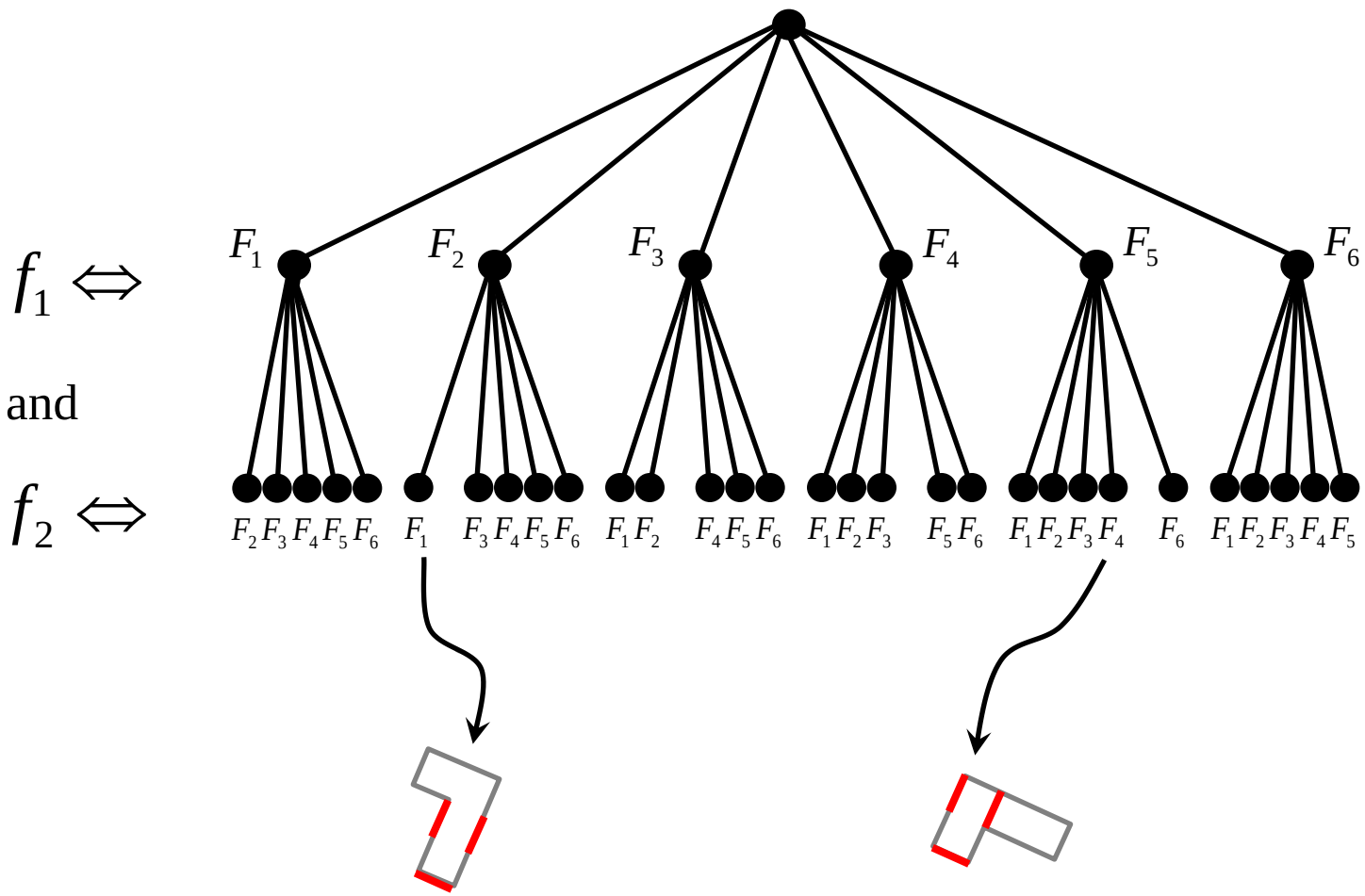
Interpretation Trees for Feature-Based Identification

The Interpretation Tree



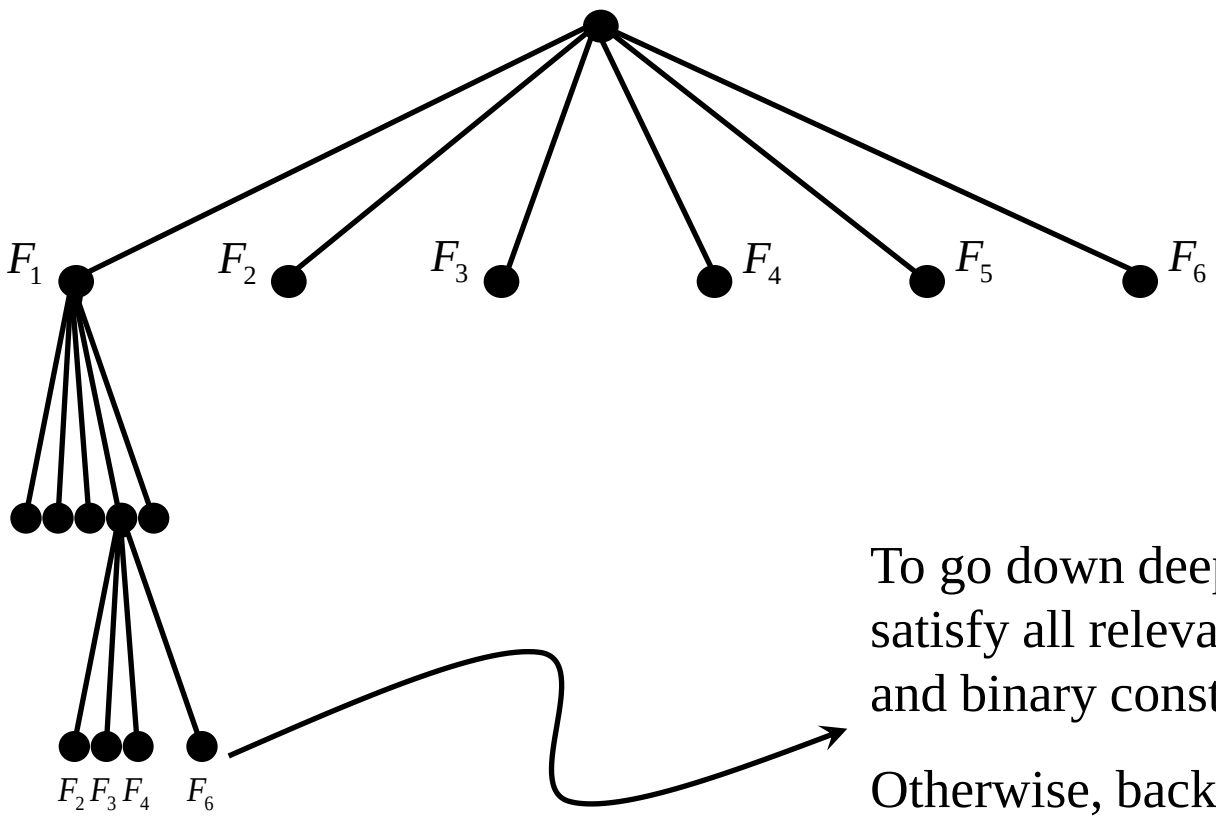
Interpretation Trees for Feature-Based Identification

The Interpretation Tree



Interpretation Trees for Feature-Based Identification

Constraint-based DFS of the interpretation tree

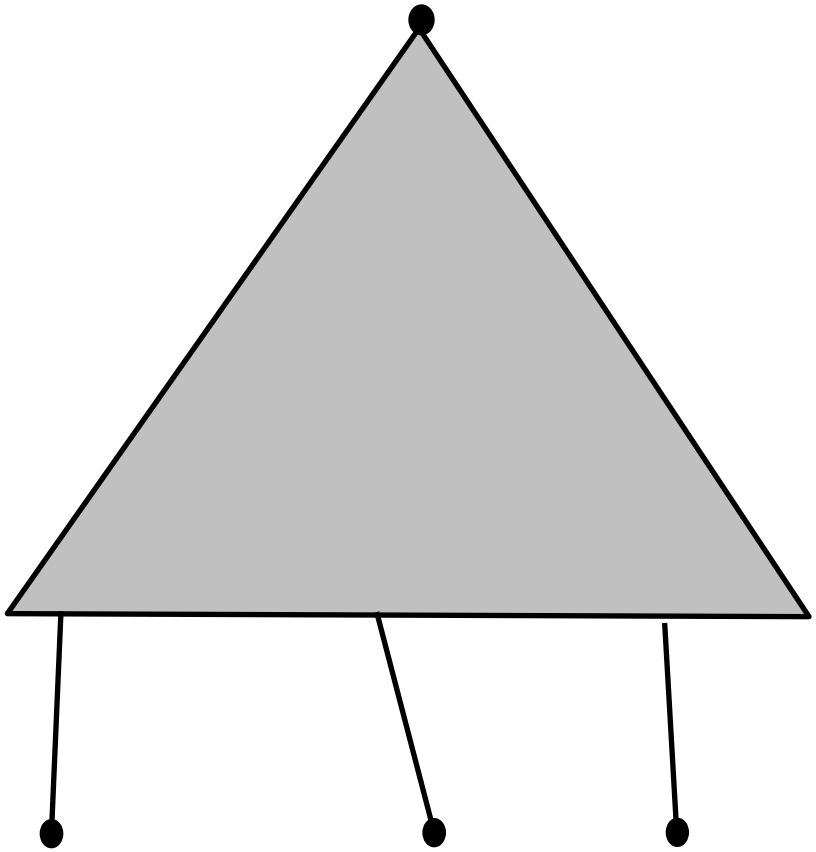


To go down deeper must satisfy all relevant unary and binary constraints.

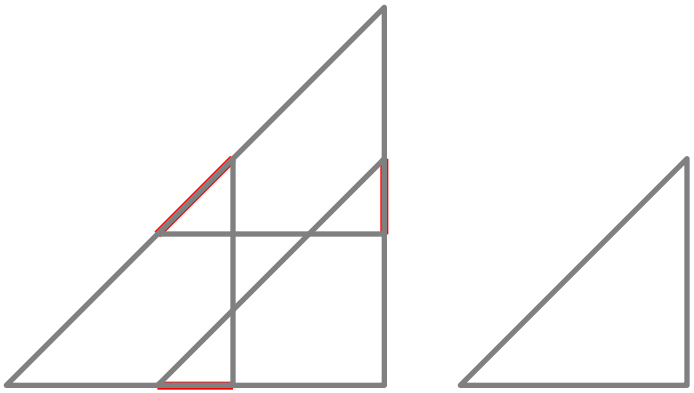
Otherwise, backtrack.

Interpretation Trees for Feature-Based Identification

Verification of hypothetical interpretations



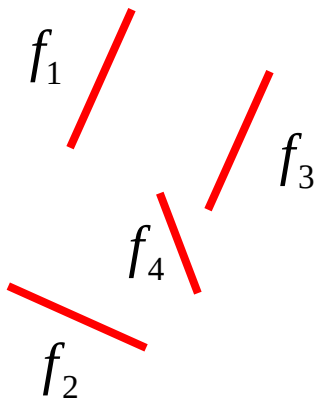
Verification



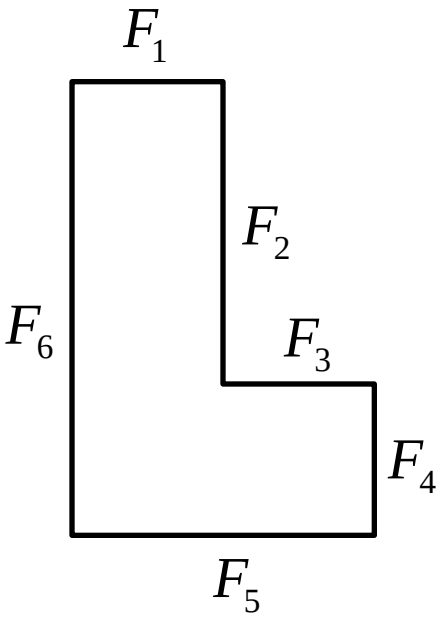
Interpretation Trees for Feature-Based Identification

Partial, misleading, or spurious features

Measurements (Image)

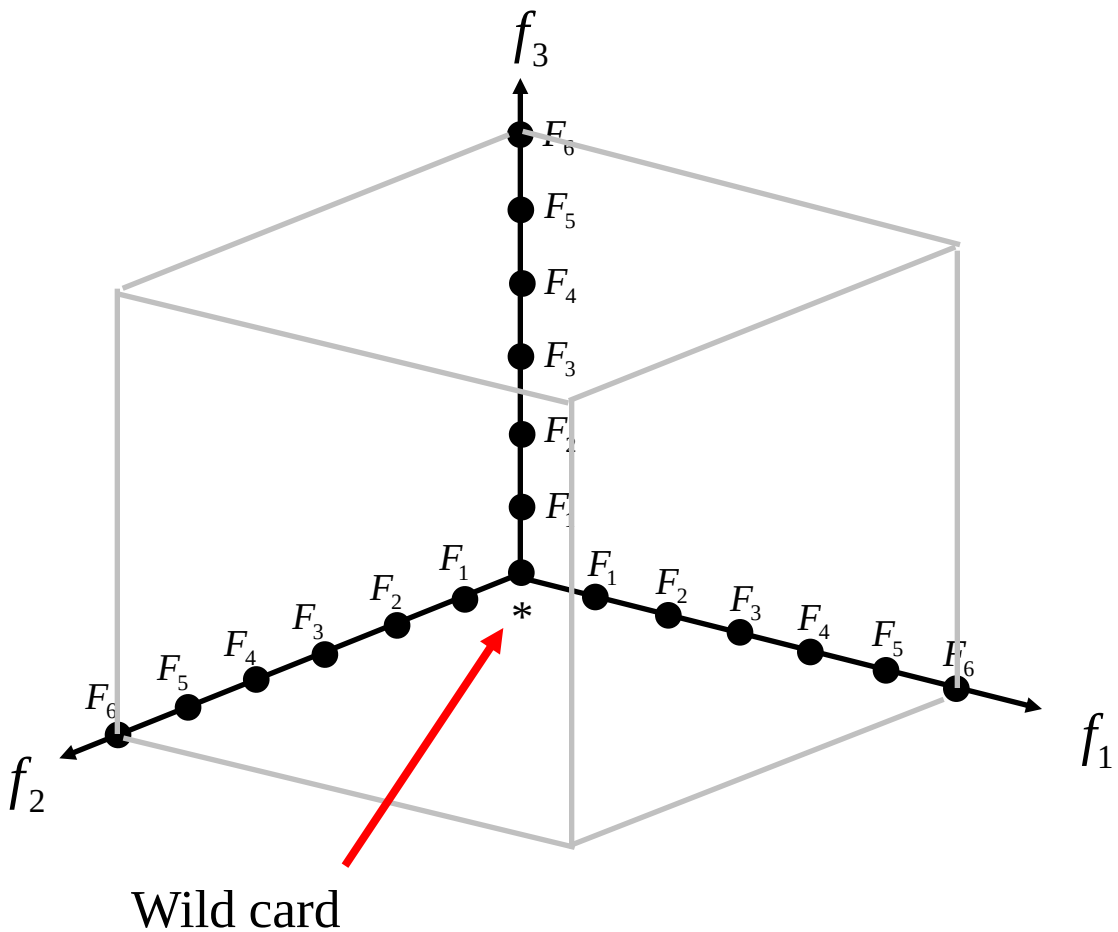


Model



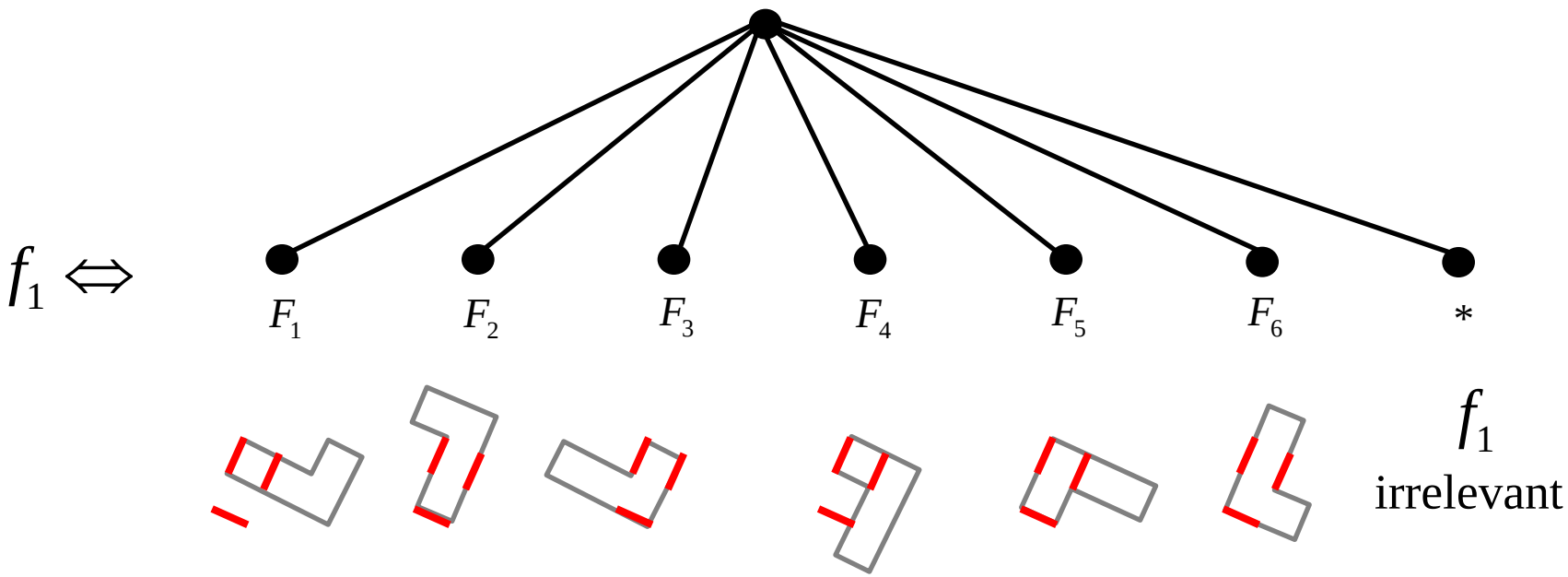
Interpretation Trees for Feature-Based Identification

The wild card



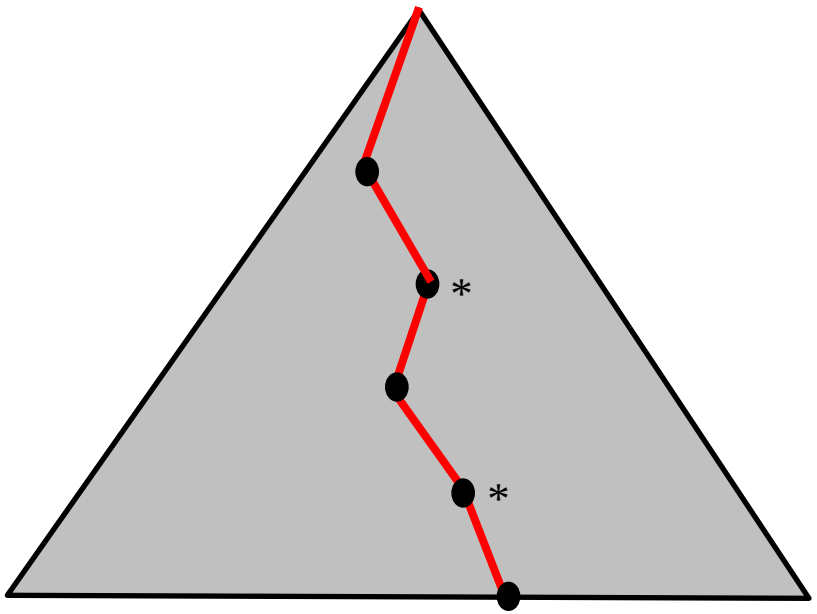
Interpretation Trees for Feature-Based Identification

The wild card



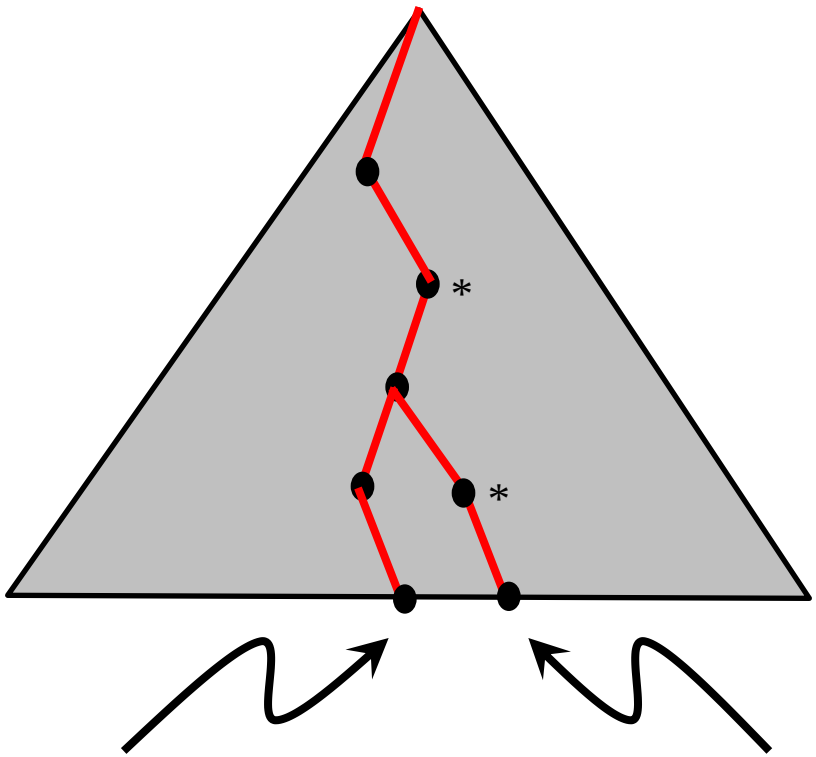
Interpretation Trees for Feature-Based Identification

Partial, misleading, or spurious features



Interpretation Trees for Feature-Based Identification

Allowing wild cards = redundant interpretations



$$\sum_{i=0}^{c-1} \binom{c}{i} = 2^c - 1$$

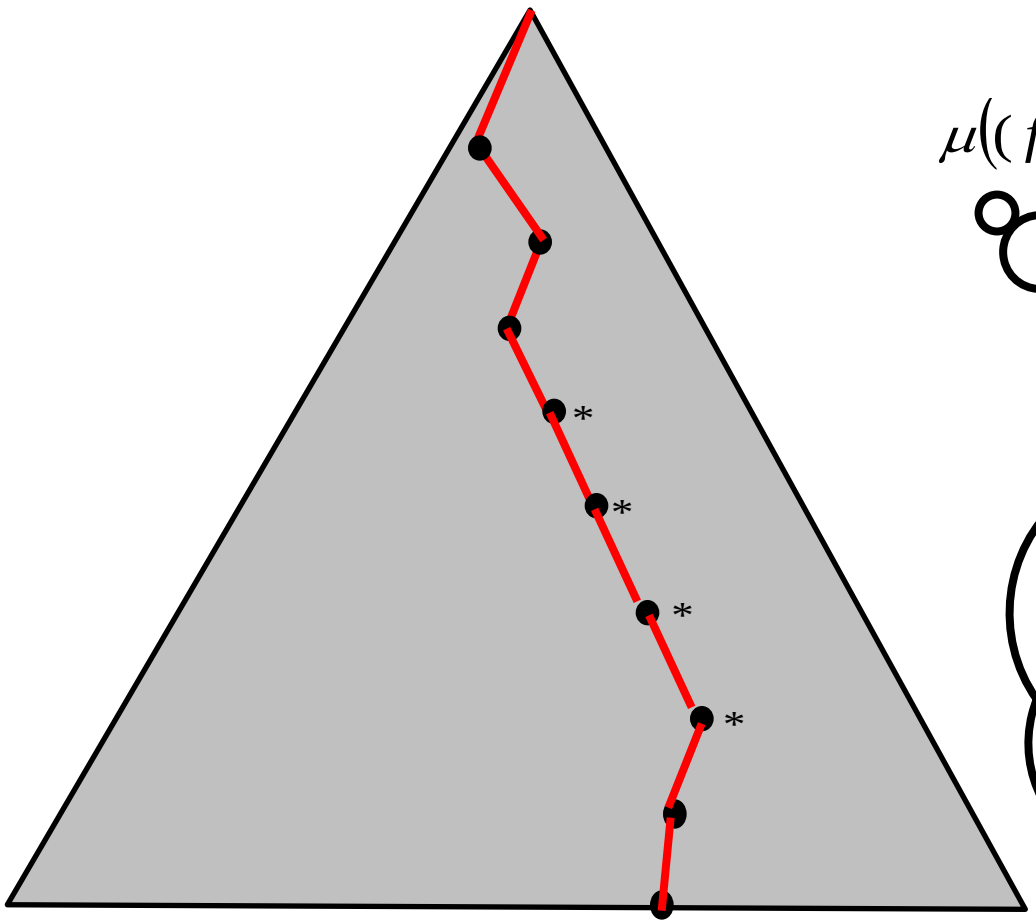
redundant interpretations

c-interpretation

(c-1)-interpretation

Interpretation Trees for Feature-Based Identification

Branch and bound heuristics



$$\mu\left((f_1, F_{j_1}); (f_2, F_{j_2}); \ldots; (f_s, F_{j_s})\right)$$

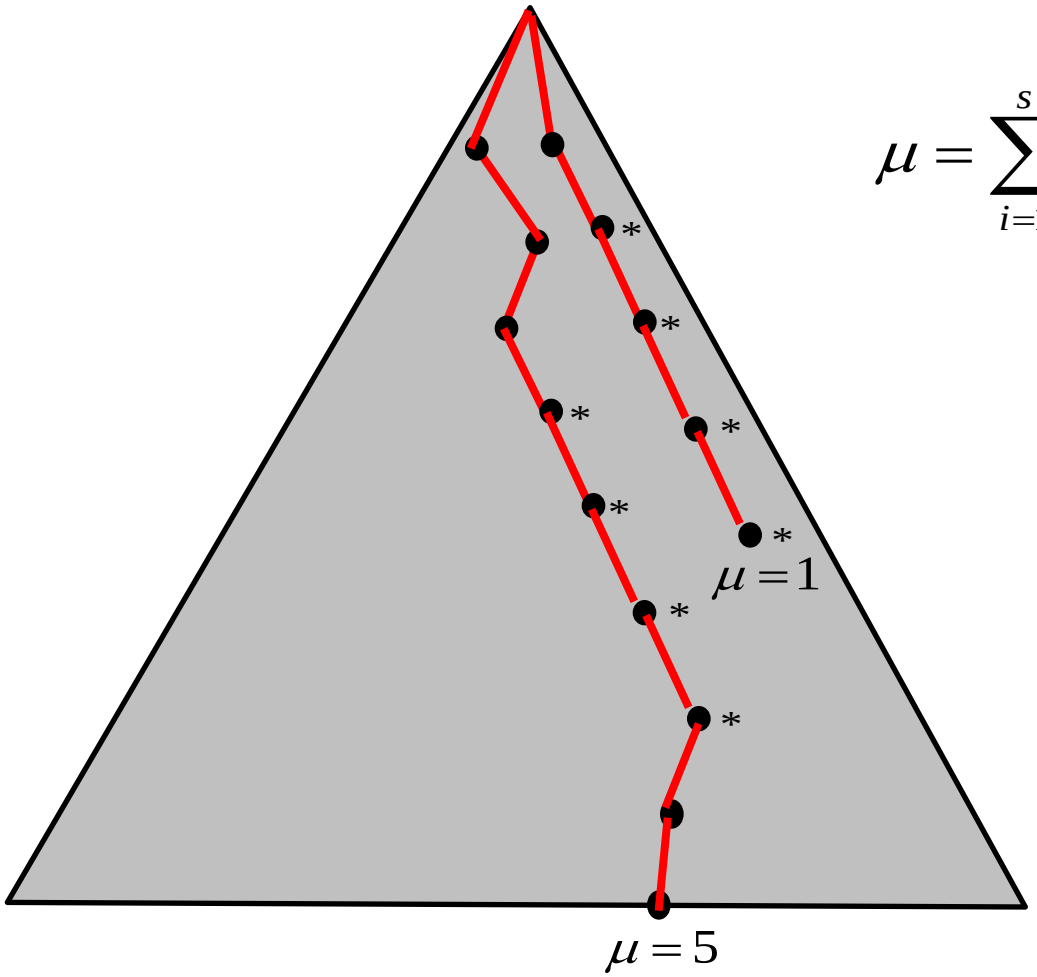
$$\mu = \sum_{i=1}^{s} \delta_i w_i$$

$$\delta_i = \begin{cases} 1 & F_{j_i} \neq * \\ 0 & otherwise \end{cases}$$

w_i = Match score

Interpretation Trees for Feature-Based Identification

Branch and bound heuristics



$$\mu = \sum_{i=1}^{s} \delta_i \qquad \delta_i = \begin{cases} 1 & F_{j_i} \neq * \\ 0 & otherwise \end{cases}$$

μ_{max}=0

At each node

Compute μ, μ_{bound}

If μ_{bound}<=μ_{max}

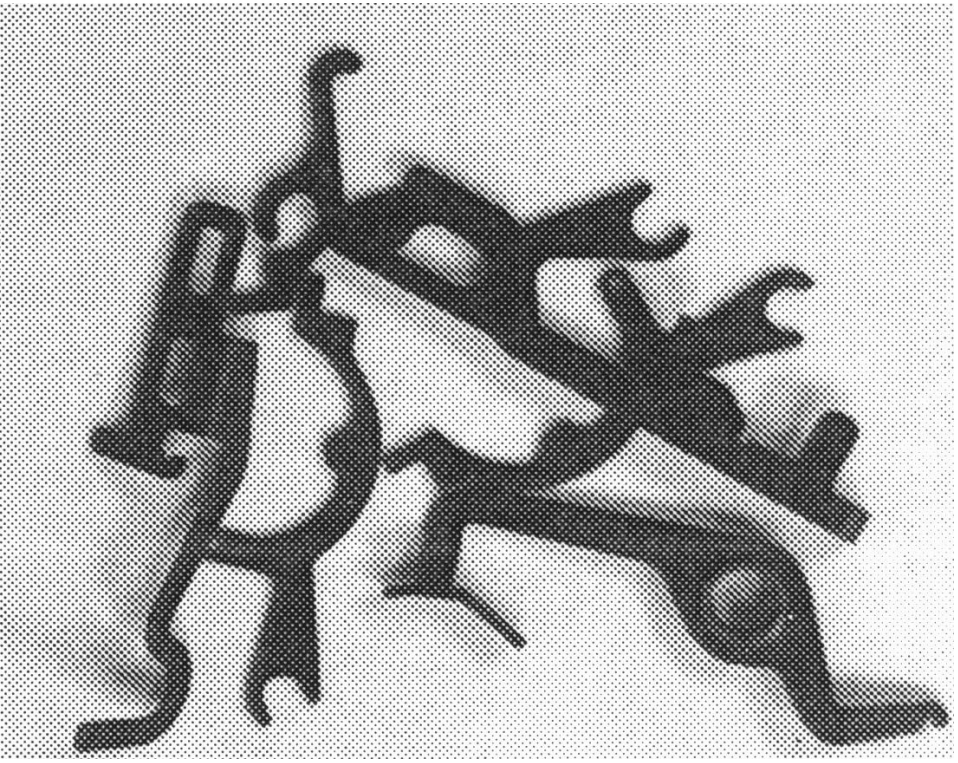
backtrack

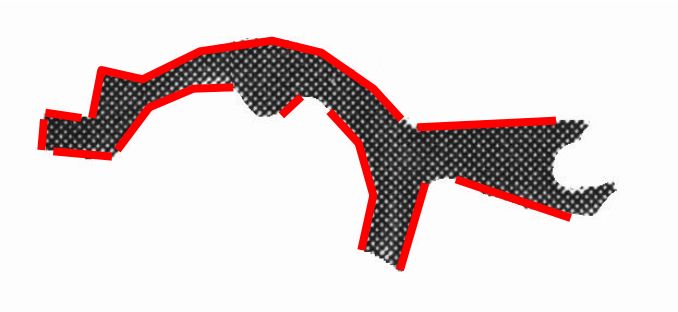
else

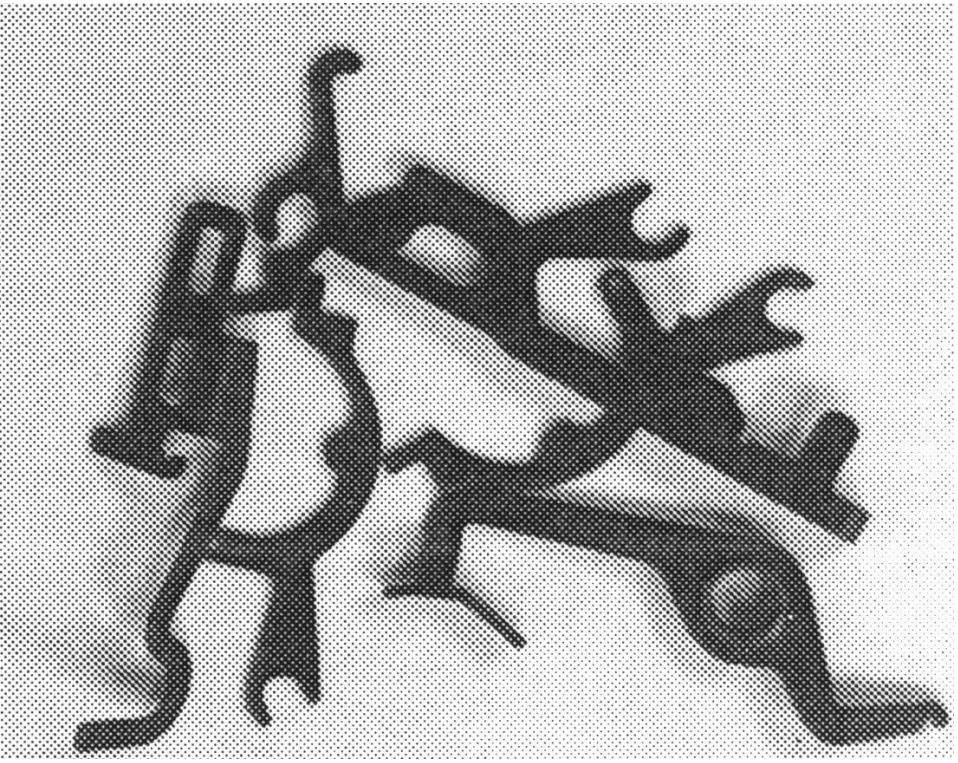
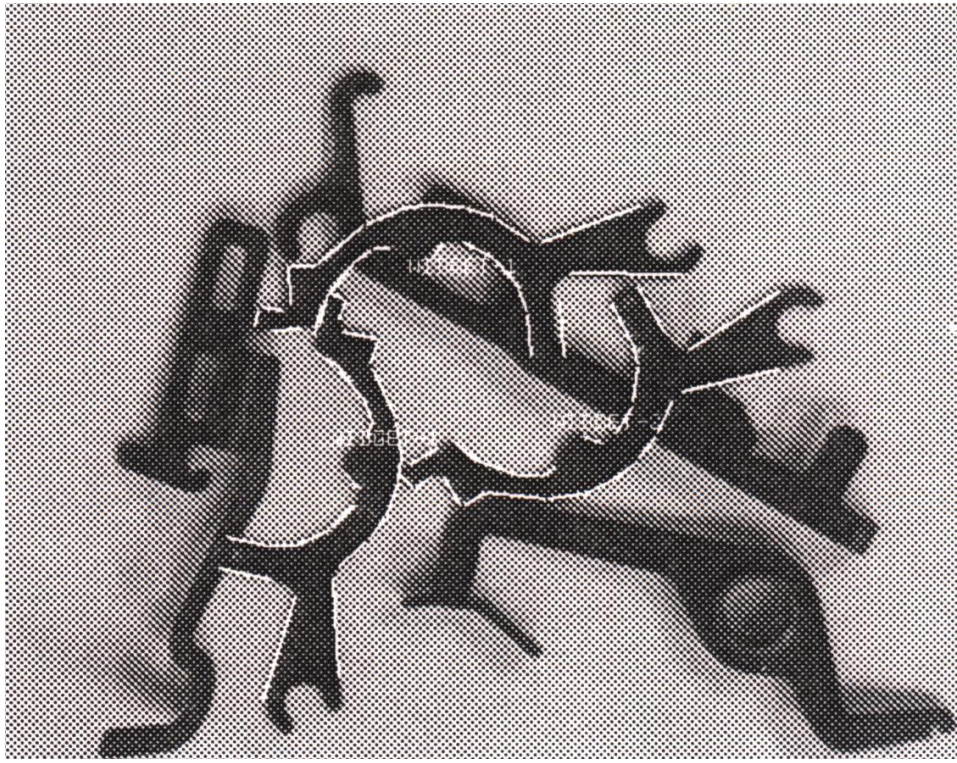
$\mu_{max=}$=μ

Interpretation Trees for Feature-Based Identification

Example





Interpretation Trees for Feature-Based Identification

Example